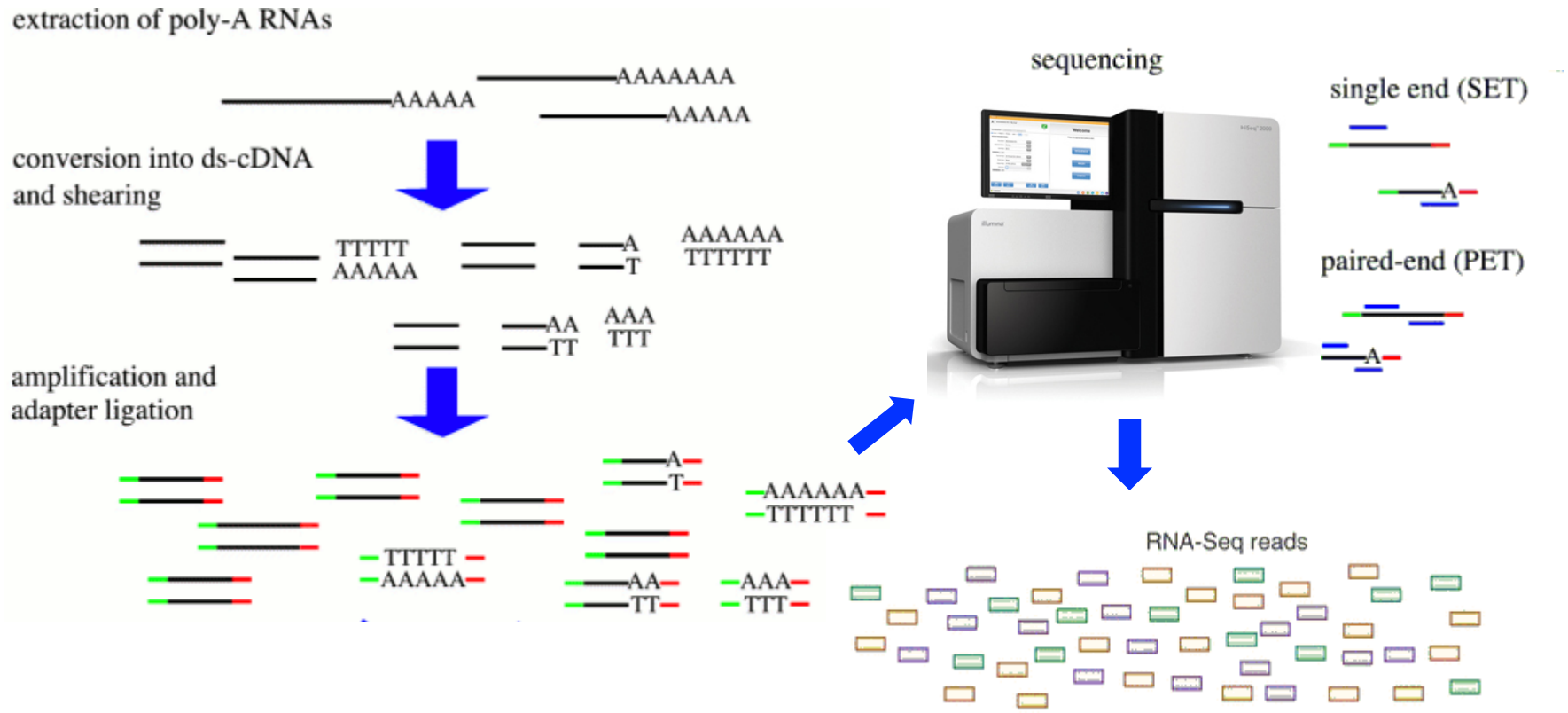
RNA-Seq data analysis protocol

Hui Yang

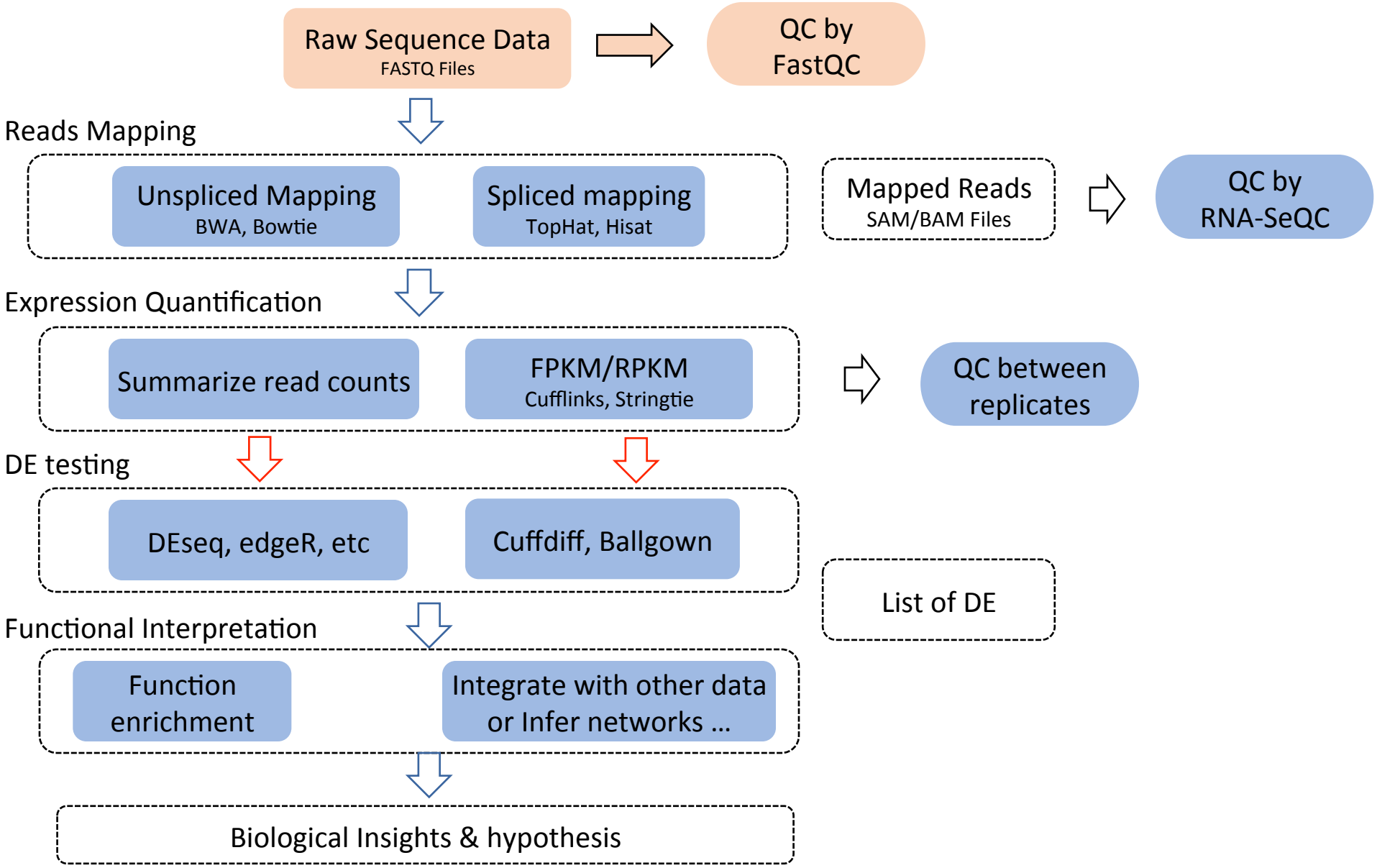
2017.12.21

Overview of RNA-Seq

Transcriptome profiling using NGS



From reads to differential expression



Obtain Raw Sequence Data

- In house RNA-Seq from Illumina *et al*

```
$ gunzip ${sample}.fastq.gz   #decompress raw file
```

- Public RNA-Seq data from GEO [GSE52194]

```
$ wget ftp:xxx/SRRxxx.sra -O ${sample}.sra  #download raw data from SRA
$ fastq-dump --split-files ${sample}.sra          #convert .sra to .fastq format
                                                  #${sample}_2.fastq for paired-end
```

➢ **Fastq format:**

Line1: Sequence identifier
Line2: Raw sequence
Line3: meaningless
Line4: quality values for the sequence

➢ **Example**

```
@tumor_rep1.40 40 length=60
CCCTCCTCATCTCCCTTGTCCTTTTCCTCTTCCTTCTCTTCTTCCAGATCGGAAGAGCGG
+tumor_rep1.40 40 length=60
CCCCCCCCCCC@CCCBBB@B>???:BBBBB>ABABBBBBBBBBBBBBBBA?6:>BBBBB
```

Raw Sequencing Data QC-FastQC

➢ **Information can be checked**

- Basic Statistics(total reads, sequence length, GC content etc.)
- Per base sequence quality
- Overrepresented sequences
- GC content
- Duplication level

  etc.

➢ **Input command line:**

  fastqc data1.fastq

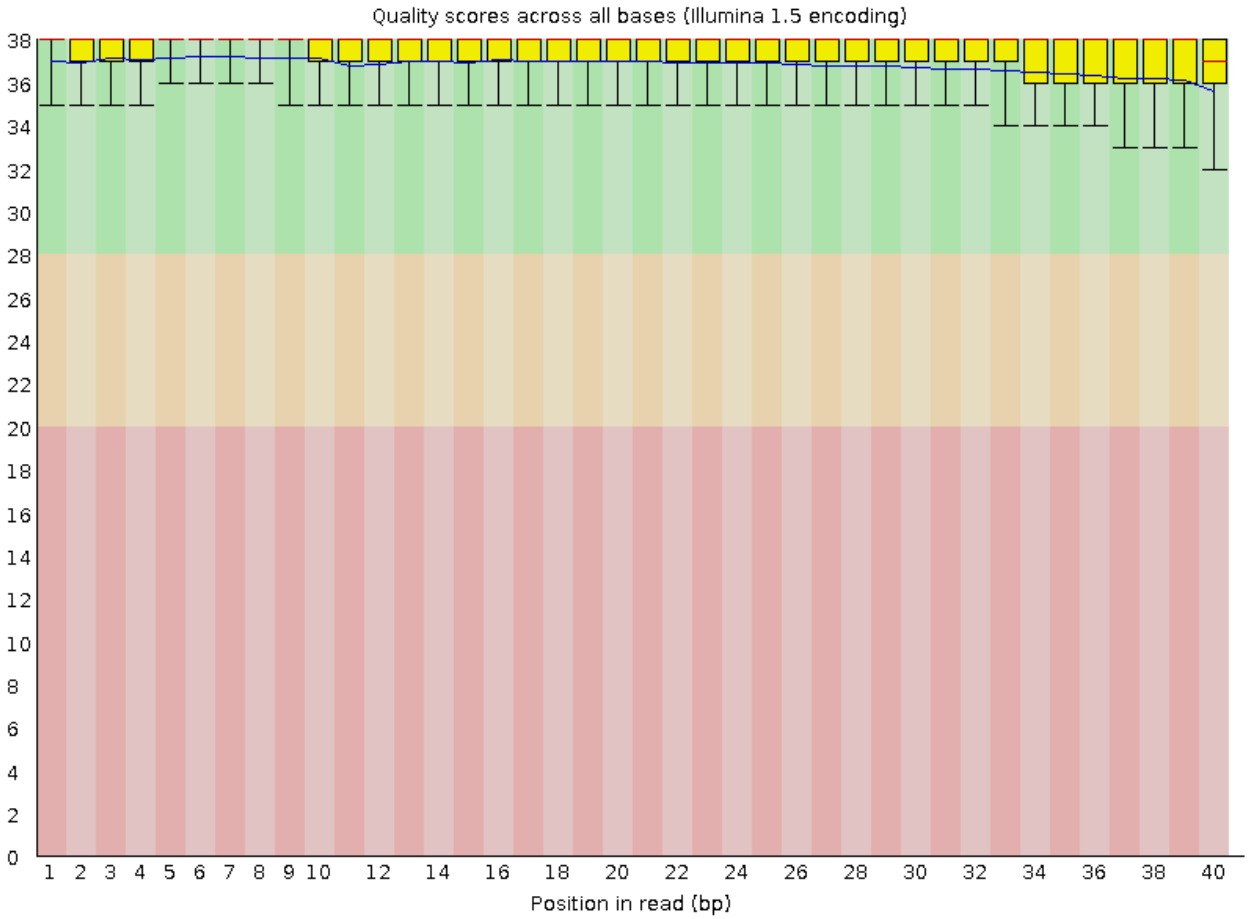
➢ **Output format**: html webpage

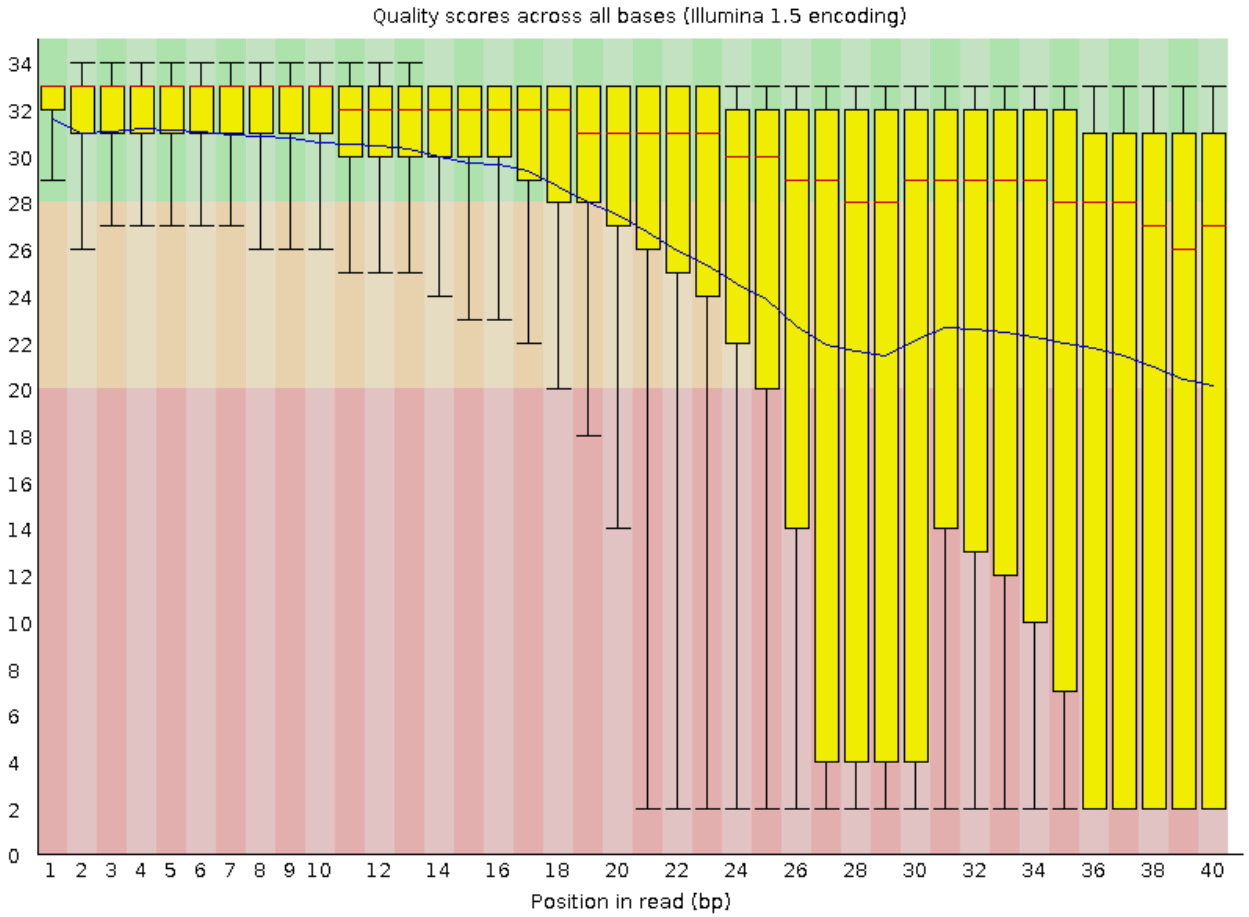
http://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Per base sequence quality





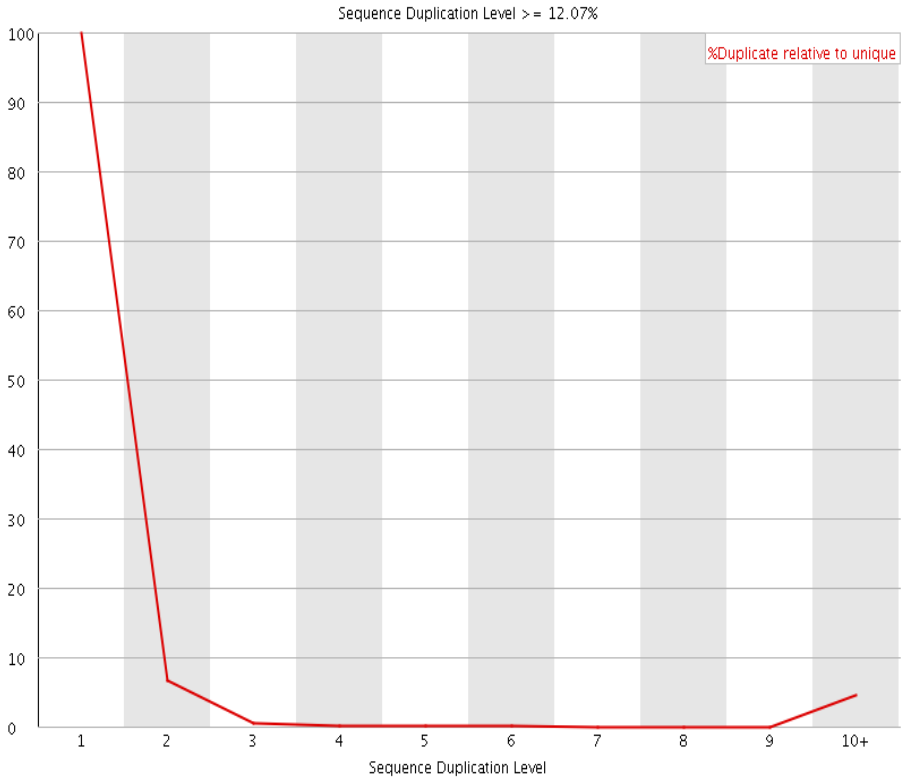


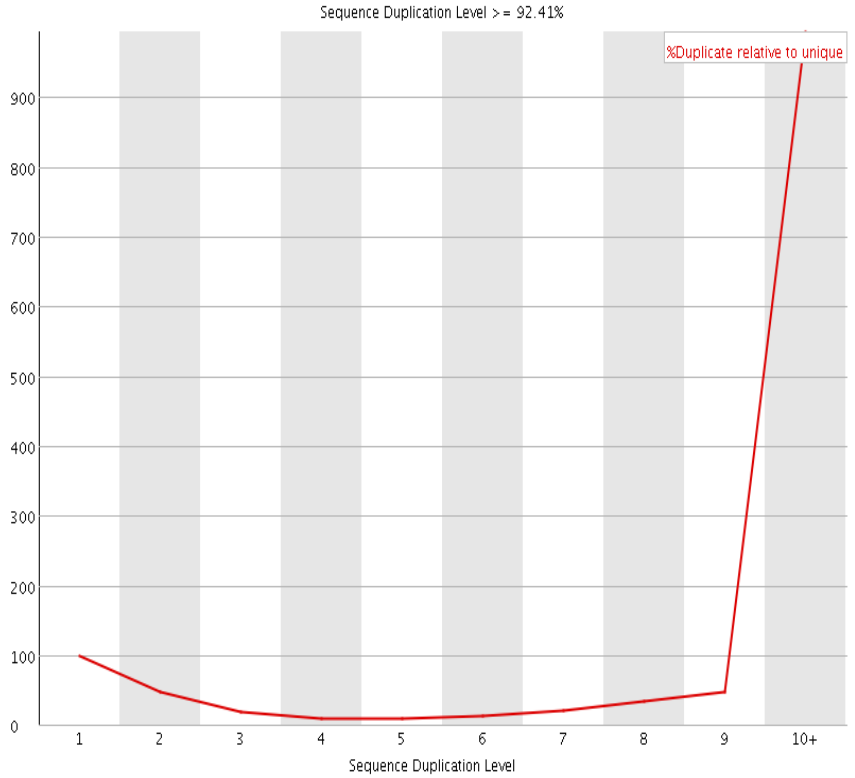


Duplication level









Overrepresented Sequences

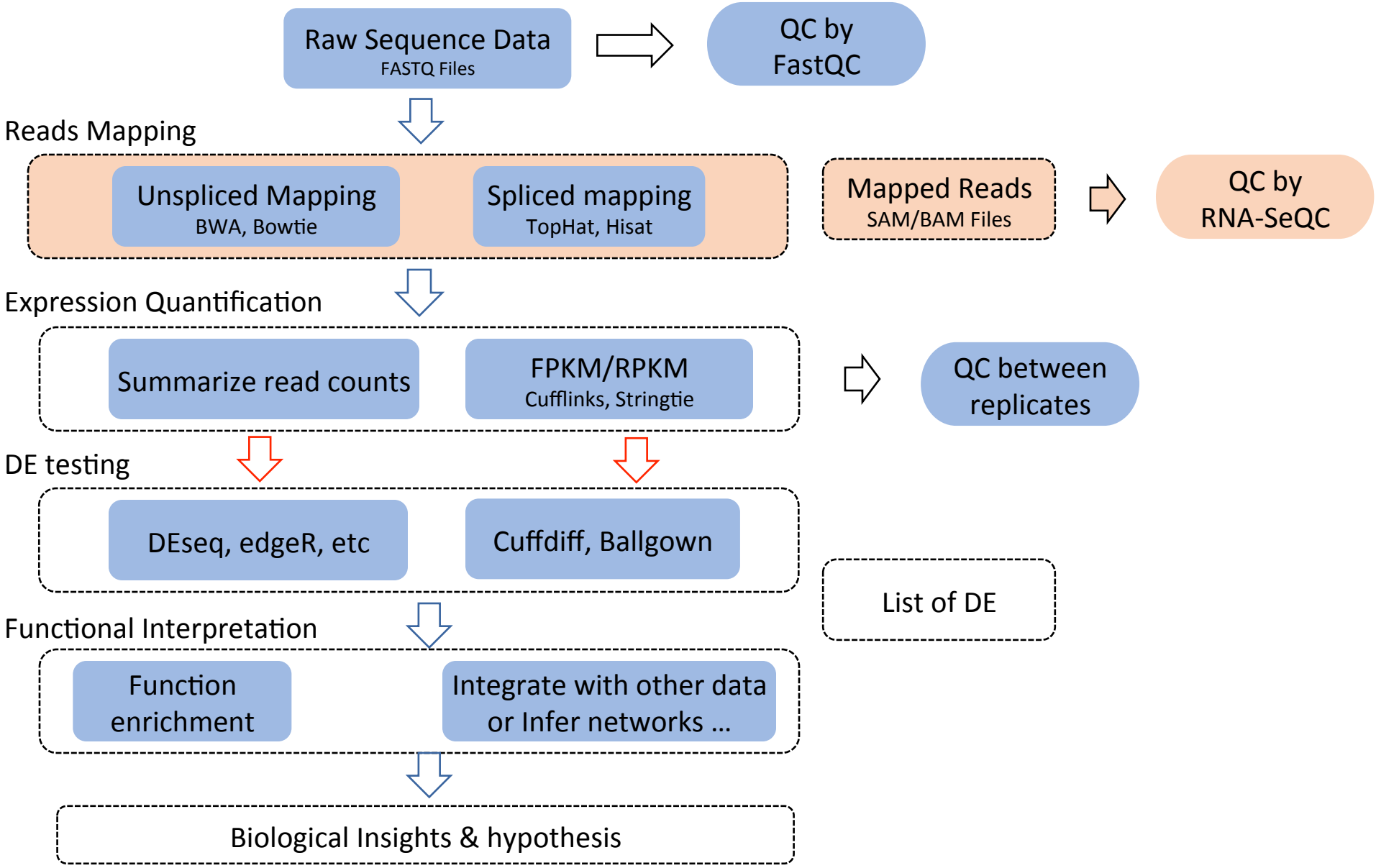
Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GTGTCAGTCACTTCCAGCGGTCGTATGCCGTCTTCT	2667259	7.236020826756234	No Hit
TATCCCCGCCTGTCACGCGGGACGTGTCAGTCACTT	703193	1.907695950497944	No Hit
CTCGCTCCTCTCCTACTTGGATAACTGGTGTCAGTC	352107	0.9552329133566171	No Hit
TGTCAGTCACTTCCAGCGGTCGTATGCCGTCTTCTG	351690	0.9541016318857297	No Hit
CTCCTCTCCTACTTGGATAACTGGTGTCAGTCACTT	247800	0.6722579100380558	No Hit
CATCATATGGTGACCTCCCGCGTGTCAGTCACTTCC	192614	0.5225435233416872	No Hit
CATCAATATGGTGACCTCCCGCGTGTCAGTCACTTC	192513	0.5222695199158848	No Hit
CATCAATATGGTGACCTCCCGGAACGTGTCAGTCAC	191604	0.5198034890836628	No Hit
CATCAATATGGTGACCTCCCCGTGTCAGTCACTTCC	163498	0.4435545753648186	No Hit
CATCATATGGTGACCTCCCCGTGTCAGTCACTTCCA	158547	0.43012298169008734	No Hit
TATCCCCGCCTCACGCGGGACGTGTCAGTCACTTCC	131347	0.3563319600878471	No Hit
AAAACGTGTCAGTCACTTCCAGCGGTCGTATGCCGT	127345	0.34547491345357634	No Hit
CATGAGACTCTTAATCTCACGTGTCAGTCACTTCCA	109695	0.29759213656829914	No Hit

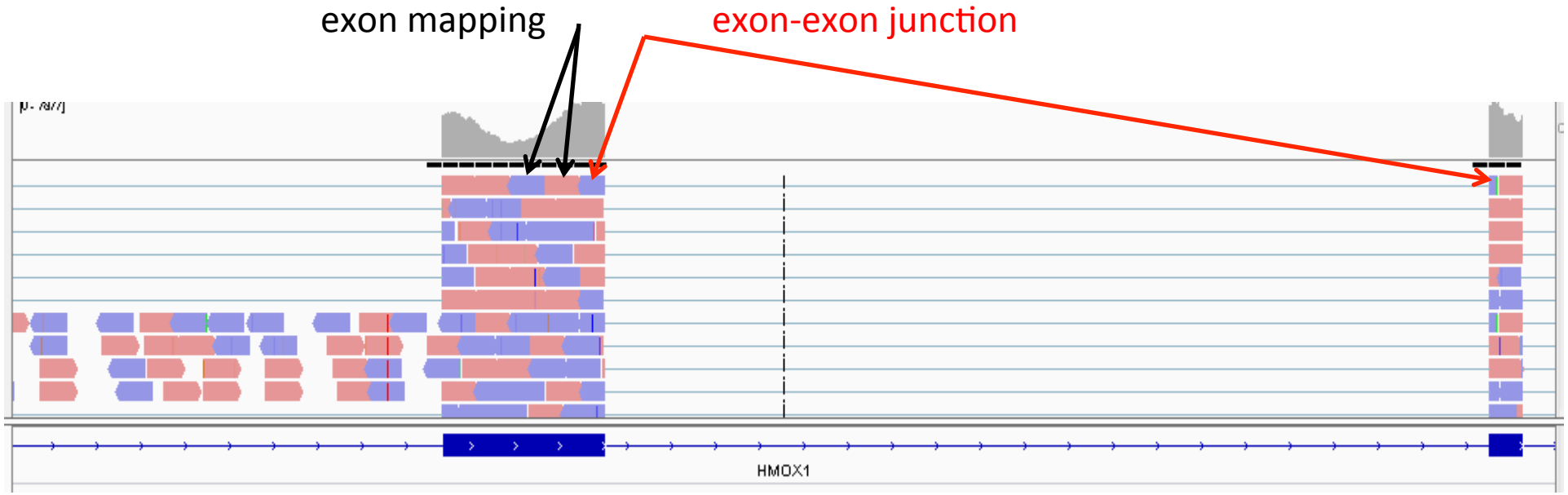
Adapter



From reads to differential expression



Read mapping



Unlike DNA-Seq, when mapping RNA-Seq reads back to reference genome, we need to pay attention to **exon-exon junction reads**

List of mapping methods

Class	Tools	Notes	Uses
Unspliced aligners	Bowtie	Index reference sequence	Aligning reads to a reference transcriptome
	BWA	Incorporates quality scores	
Spliced aligners	Tophat	Use Bowtie alignments	Aligning reads to a reference genome. Allows for the identification of novel splice junctions
	Hisat	Use a graph FM index	
	STAR	Use sequential maximum mappable seed search	

Usual mapping methods-tophat2

- **Command line usage**:

  tophat2 [options]* <genome_index_base> <reads1_1[,...,readsN_1]> [reads1_2,...readsN_2]

- **Typical Parameter**:

<genome_index_base>	The basename of the genome
<reads1_1[,...,readsN_1]>	Reads files, *_1 format when paired-end
[reads1_2,…readsN_2]	Only used when paired-end, the order must be the same as *_1 files.
-r/--mate-inner-dist <int>	Expected inner distance between mate pairs
--mate-std-dev <int>	Standard deviation for the distribution on inner distances
--no-novel-juncs	Only look for reads across junctions indicated in supplied GFF.

- Example:

```
$ tophat2 -p 8 --mate-inner-dist 0 --mate-std-dev 20 -o ./ -G ${gtf} ${bowtieIndex} ${sample}_1.fastq ${sample}_2.fastq
```

Usual mapping methods-hisat2

- **Command line usage**:

  hisat2 [options]* -x <ht2-idx> {-1 <m1> -2 <m2> | -U <r>} [-S <sam>]

- **Typical Parameter**:

  <ht2-idx> Index filename prefix (minus trailing .X.ht2)

  <m1> Files with #1 mates, paired with files in <m2>

  <m2> Files with #2 mates, paired with files in <m1>

  <r> Files with unpaired reads

  <sam> File for SAM output

- Example:

```
$ hisat2 -p 8 --dta -x ${hisat2Index} -1 ${sample}_1.fastq -2 ${sample}
  _2.fastq -S ${sample}.sam --known-splicesite-infile ${ss}
```

- **generate sorted bam which is needed for stringtie or cufflinks**
  - samtools sort -@ 8 -o accepted_hits.bam accepted_hits.sam
  - -@ : thread number

Mapping QC

- Percentage of reads properly mapped or uniquely mapped

  'align_summary.txt' generated after tophat and log file after hisat2

```
    9336231 pairs aligned 0 times concordantly or discordantly; of these:
      18672462 mates make up the pairs; of these:
        9842822 (52.71%) aligned 0 times
        4485522 (24.02%) aligned exactly 1 time
        4344118 (23.26%) aligned >1 times
88.92% overall alignment rate
```

  - **Expection**: not less than 60%, more higher is better(>80%)

- Reads distribution by read_distribution in RNA-SeQC(http://rseqc.sourceforge.net/)

  Command: read_distribution.py -i accepted_hits.bam -r ${refseq} > dirstribution.txt

tumor_rep1.distribution

Total Reads	79026622		
Total Tags	89793605		
Total Assigned Tags	75809521		
==========	==========	==========	==========
Group	Total_bases	Tag_count	Tags/Kb
CDS_Exons	35120052	13494515	384.24
5'UTR_Exons	11280508	20944611	1856.71
3'UTR_Exons	32501976	5423019	166.85
Introns	1220794231	15932040	13.05
TSS_up_1kb	19171505	680327	35.49
TSS_up_5kb	86869776	3107567	35.77
TSS_up_10kb	159692708	4124535	25.83
TES_down_1kb	20028908	559390	27.93
TES_down_5kb	86466769	14102441	163.10
TES_down_10kb	154361762	15890801	102.95
==========	==========	==========	==========

Output:

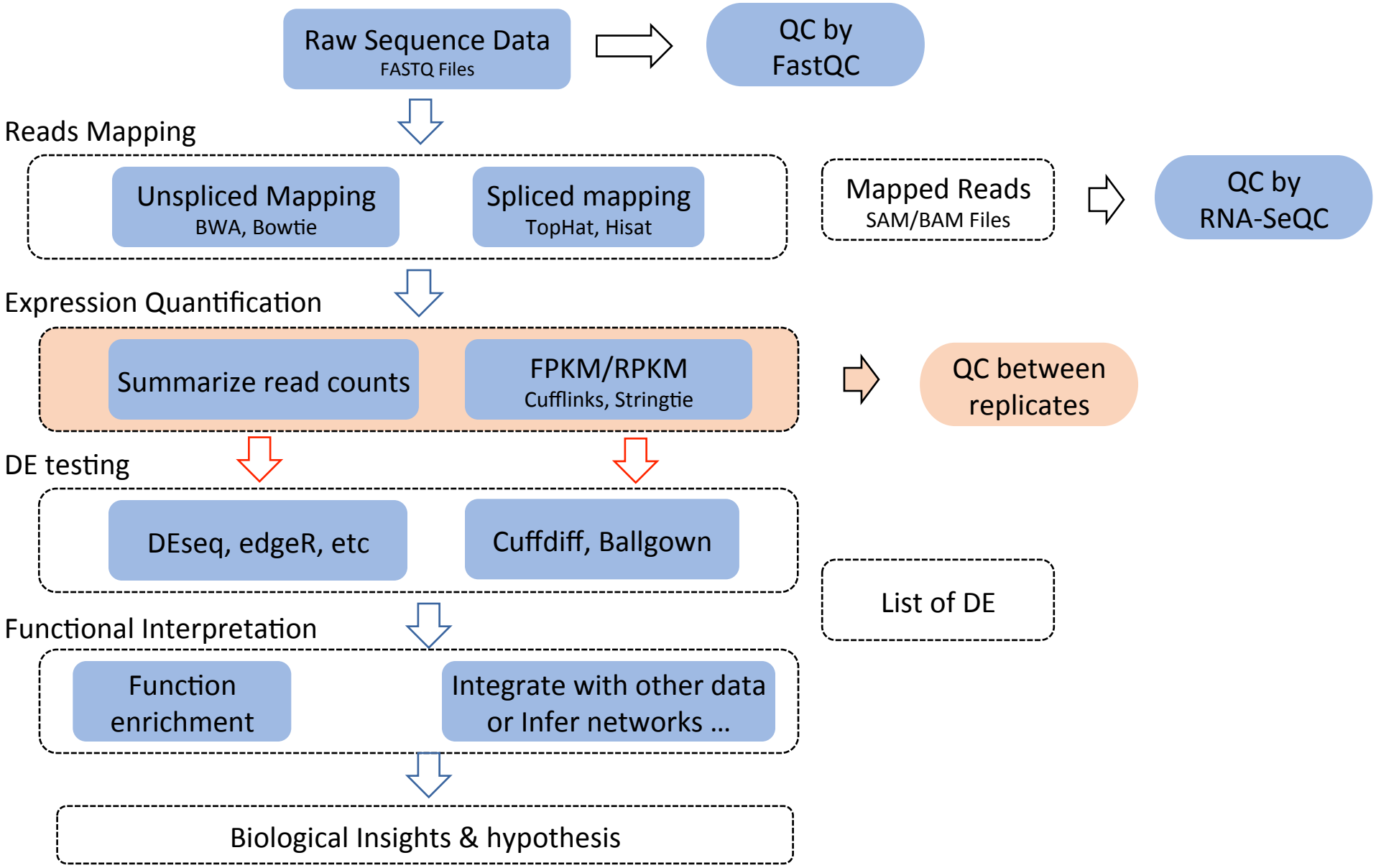
Exon percentage = Exon Tags / Total Tags

Intron percentage = Introns Tag_count/Total Tages

Expection:

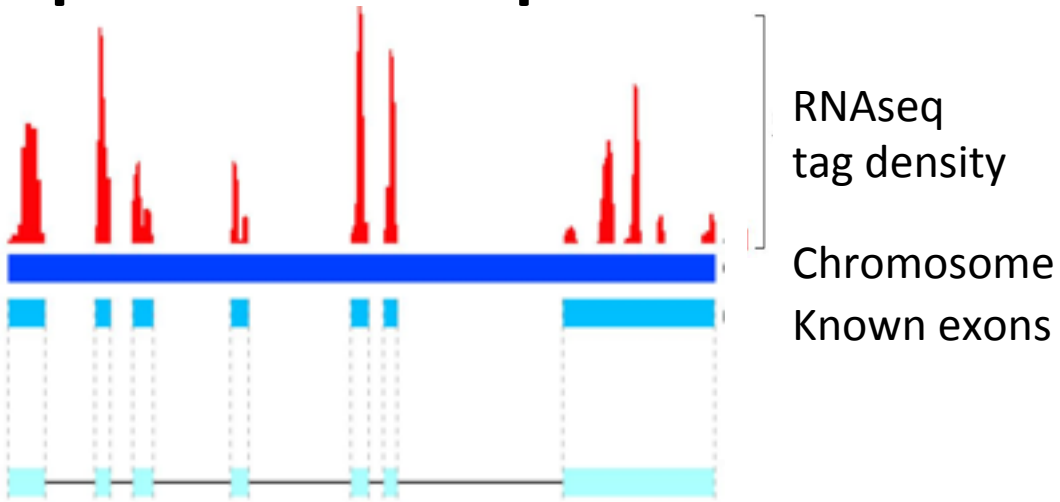
CDS percentage is more high such as >40% while Intron percentage is low, such as no more than 20%.

From reads to differential expression



Expression quantification

Count data



- htseq-count
  - **Command**: htseq-count [options] alignment_file gff_file
  - **Parameters**:

    -f SMAPLE: type of <alignment_file> data, either 'sam' or 'bam'
    -r ORDER: sorting order of <alignment_file> by 'name' or 'pos'
    -i IDATTR: GFF attribute to be used as feature ID.
  - **Example**:

```
$ htseq-count -f bam accepted_hits.bam -r pos -i transcript_id ${gtf} > ${sample}.count
$
```

Expression quantification

The number of reads is roughly proportional to

- the length of the gene
- the total number of reads in the library

- FPKM /RPKM
  - $RPKM{\downarrow}geneA = Number\ of\ reads\ mapped\ to\ geneA \times 10{\uparrow}3\ \times 10{\uparrow}6\ / Total\ number\ of\ mapped\ reads\ from\ given\ library \times gene\ length\ in\ bp$
- TPM
  - $TPM{\downarrow}geneA = Number\ of\ teads\ mapped\ to\ geneA \times Read\ length \times 10{\uparrow}6\ / Total\ number\ of\ transcripts\ sampled \times gene\ length\ in\ bp$
- usual quantification methods
  - cufflinks and stringtie

Usual quantification methods

- cufflinks
  - **Command**:
    cufflinks [options] <hits.sam>
  - **Output**:
    - transcripts.gtf: transcriptome assembly
    - isoforms.fpkm_tracking /genes.fpkm_tracking: transcript/gene level expression
  - **Example**

```
$ cufflinks -G hg19_refSeq.gtf -output-dir ./ accepted_hits.bam
```

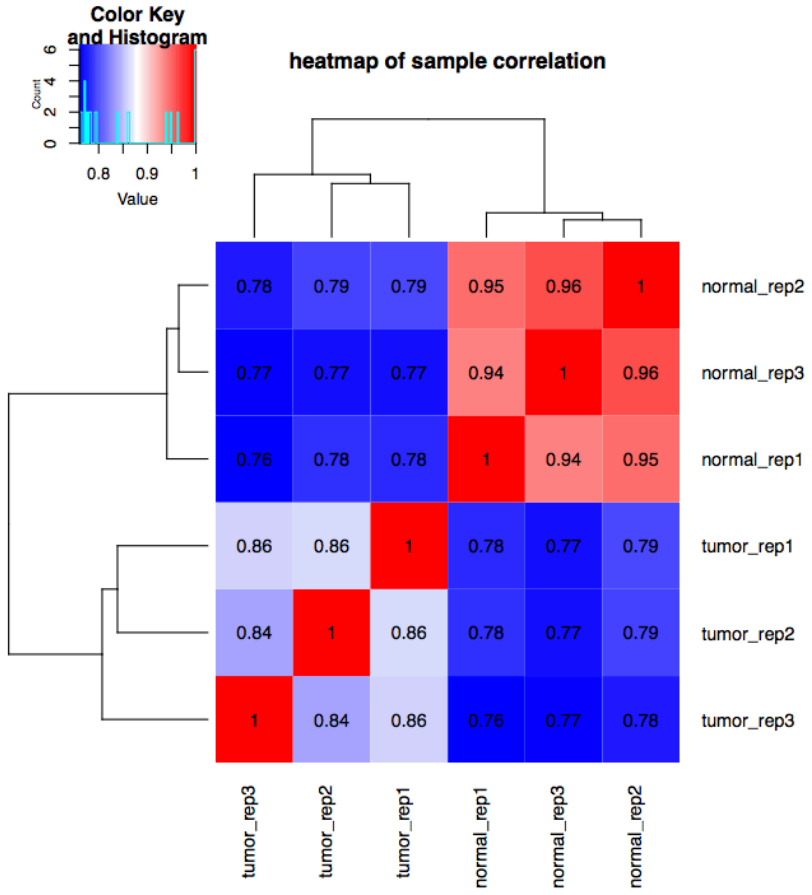
- stringtie
  - **Command**:
    stringtie <aligned_reads.bam> [options]*
  - Parameter:
    -A <gene_abund.tab> : gene abundances output file
    -B: enable the output of Ballgown input table files(*.ctab)
    -e: limit the processing to only the assembled transcripts matching the reference transcripts given with -G
  - **Output**:
    - data.gtf: GTF file containing the assembled transcripts
    - gene_abund.tab: gene abundances (contains FPKM / TPM)
    - Ballgown input table files: e2t.ctab, e_data.ctab, i2t.ctab, i_data.ctab, t_data.ctab
  - **Example**

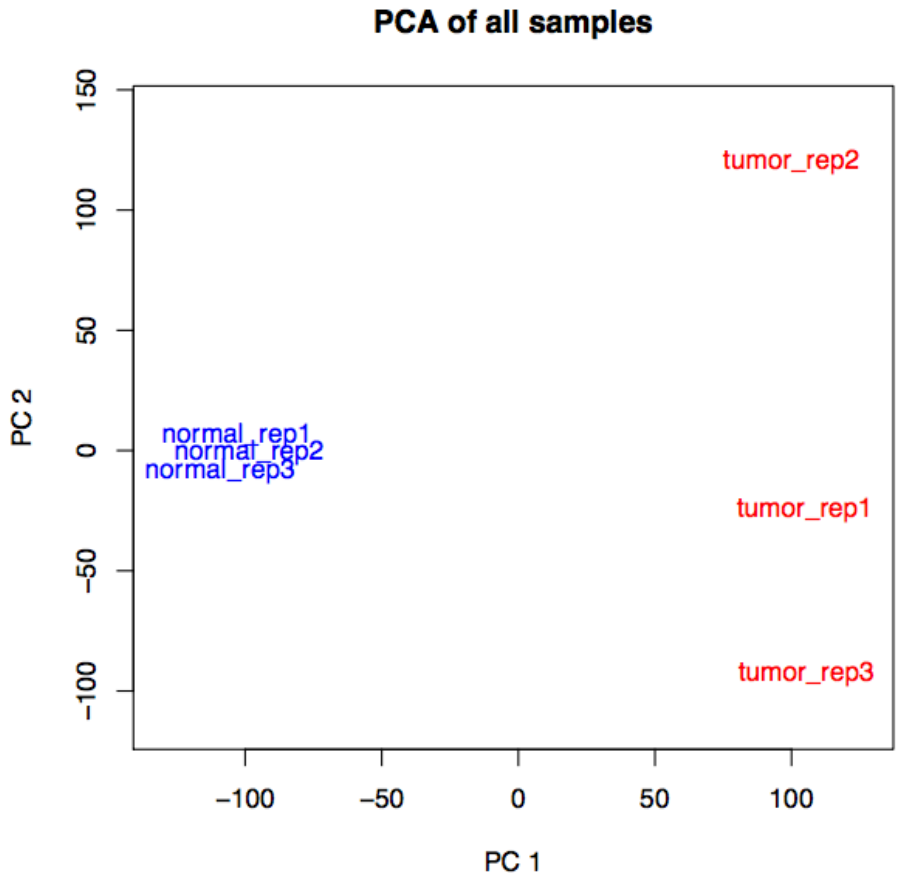
```
$ stringtie accepted_hits.bam -o ${sample}.gtf -p 8 -G ${gtf} -A gene_abund.tab -B -e
```

QC between replicates

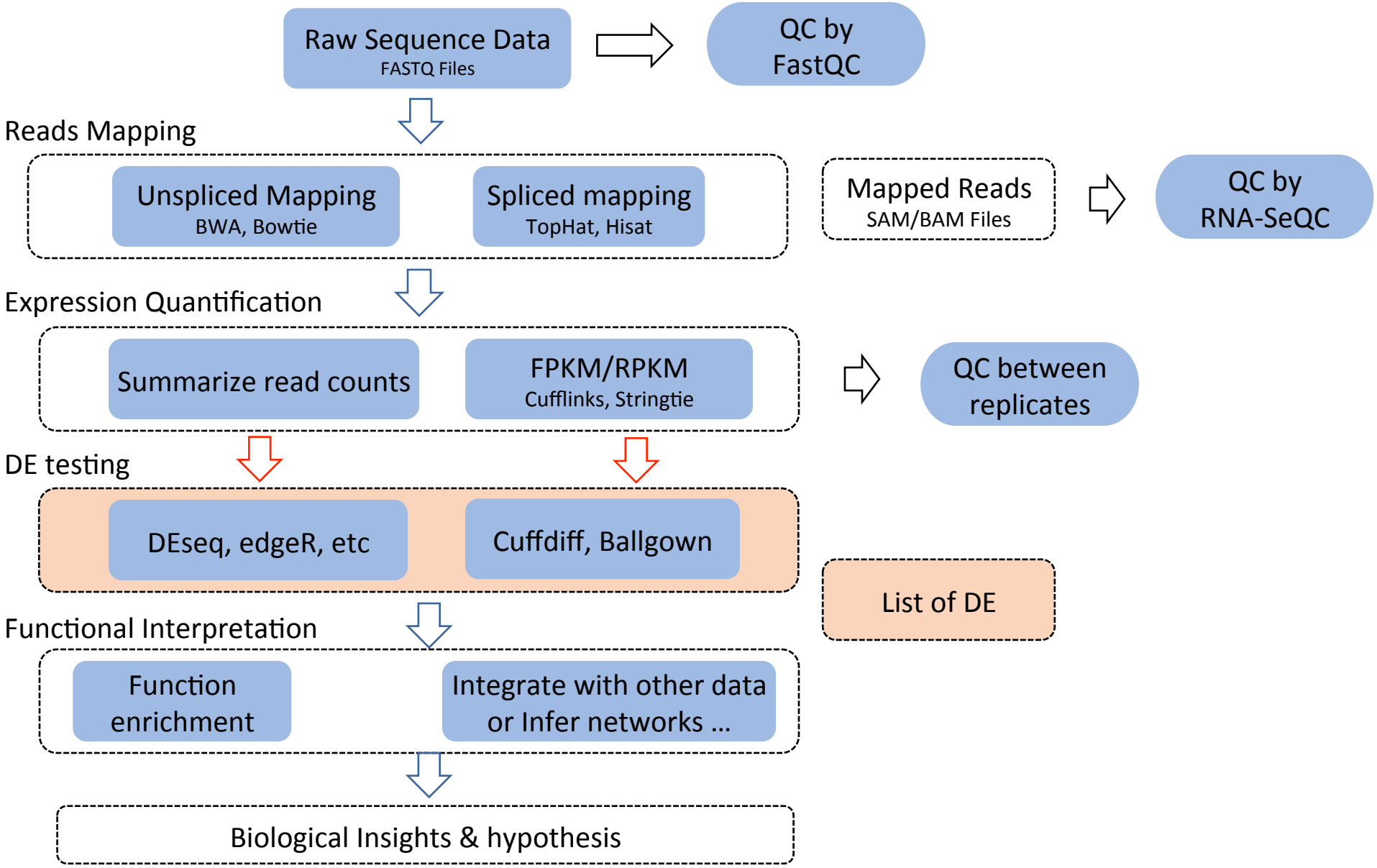
Measure the quality of samples according to biological background:

- heatmap of correlation
- PCA(principal component analysis)





From reads to differential expression



GFOLD

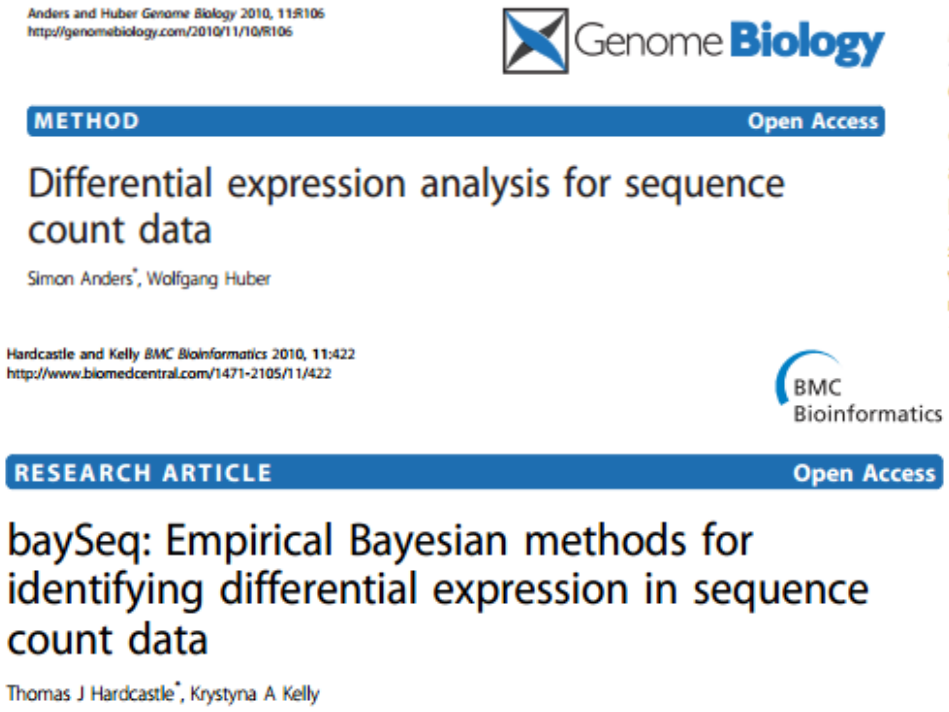
- **GFOLD** is based on the posterior distribution of log fold change.
- **Command**:

```
$ samtools view ${sample}.bam | gfold count -ann ${gtf} -tag stdin -o ${sample}.read_cnt #count reads
$ gfold diff -s1 sample1, sample2 -s2 sample3 -suf .read_cnt -o 12vs3.diff #DEGs
```

- **Output**:
  - main output file contains GFOLD value, which can be considered as a reliable log2 fold change.
  - get DEGs according to 'GFOLD' [fold change = 2]:
    - if GFOLD > 1, upregulated
    - if GFOLD < -1, downregulated
- **GFOLD** is especially useful when no replicate is available.

Count-based methods (R packages)

1. edgeR -- use an overdispersed Poisson model
2. **DESeq** -- based on negative binomial distribution
3. **baySeq** -- use an empirical Bayes approach


Anders and Huber Genome Biology 2010, 11:R106
http://genomebiology.com/2010/11/10/R106

Genome Biology

METHOD Open Access

Differential expression analysis for sequence count data

Simon Anders*, Wolfgang Huber

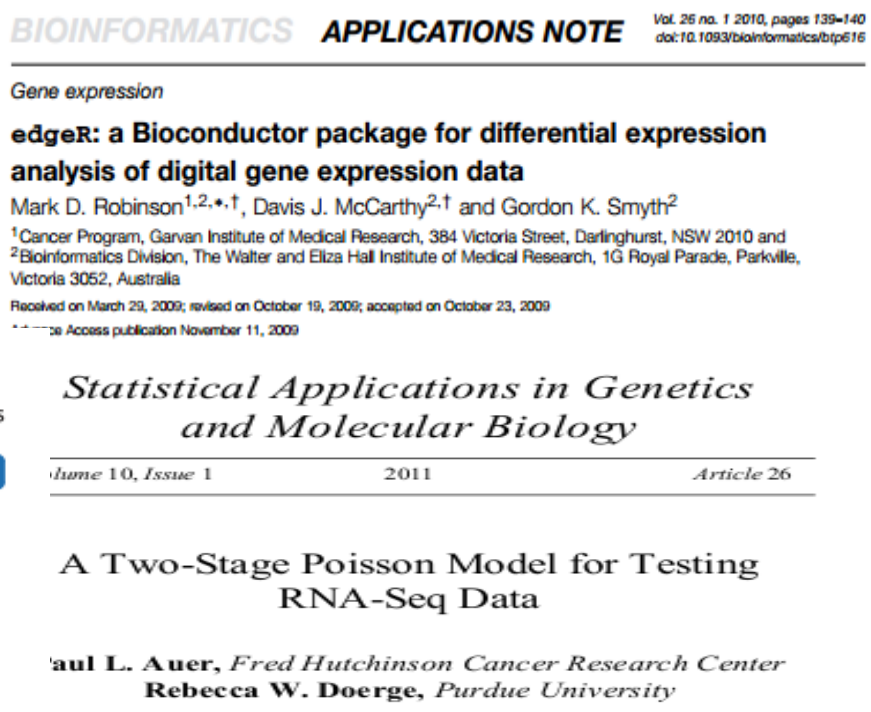
Hardcastle and Kelly BMC Bioinformatics 2010, 11:422
http://www.biomedcentral.com/1471-2105/11/422

BMC Bioinformatics

RESEARCH ARTICLE Open Access

baySeq: Empirical Bayesian methods for identifying differential expression in sequence count data

Thomas J Hardcastle*, Krystyna A Kelly


BIOINFORMATICS APPLICATIONS NOTE Vol. 26 no. 1 2010, pages 139–140
doi:10.1093/bioinformatics/btp616

Gene expression

edgeR: a Bioconductor package for differential expression analysis of digital gene expression data

Mark D. Robinson[1,2,*,†], Davis J. McCarthy[2,†] and Gordon K. Smyth[2]

[1]Cancer Program, Garvan Institute of Medical Research, 384 Victoria Street, Darlinghurst, NSW 2010 and [2]Bioinformatics Division, The Walter and Eliza Hall Institute of Medical Research, 1G Royal Parade, Parkville, Victoria 3052, Australia

Received on March 29, 2009; revised on October 19, 2009; accepted on October 23, 2009

Access publication November 11, 2009

Statistical Applications in Genetics and Molecular Biology

lume 10, Issue 1 2011 Article 26

A Two-Stage Poisson Model for Testing RNA-Seq Data

aul L. Auer, *Fred Hutchinson Cancer Research Center*
Rebecca W. Doerge, *Purdue University*

RPKM/FPKM-based methods

- Cuffdiff
  - **Command**:

```
$  cuffdiff [options] <transcripts.gtf> <sample1_hits.sam> <sample2_hits.sam> [… sampleN_hits.sam]
$
```

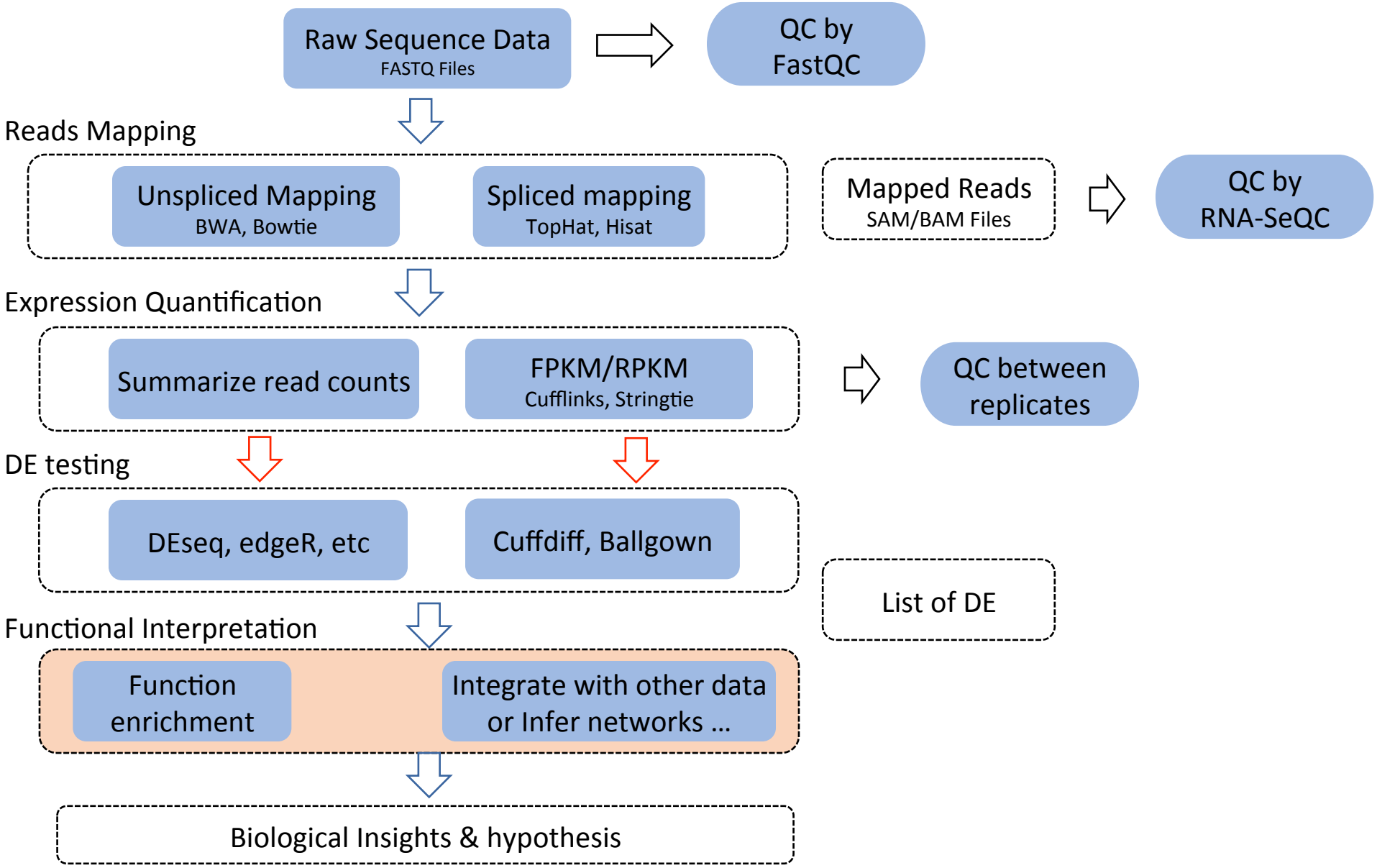
  - **Output**:

    isoform_exp.diff, gene_exp.diff, tss_group_exp.diff, cds_exp.diff

    #significant changes for spliced transcripts, primary transcripts, genes and coding sequences.

- ballgown (R packages)
  - based on basic linear modeling strategy

From reads to differential expression



Functional Interpretation

- Functional enrichment of DEGs using DAVID(https://david.ncifcrf.gov/summary.jsp)

